



Predictive Modeling of Rootzone Variables Using Greenhouse Data

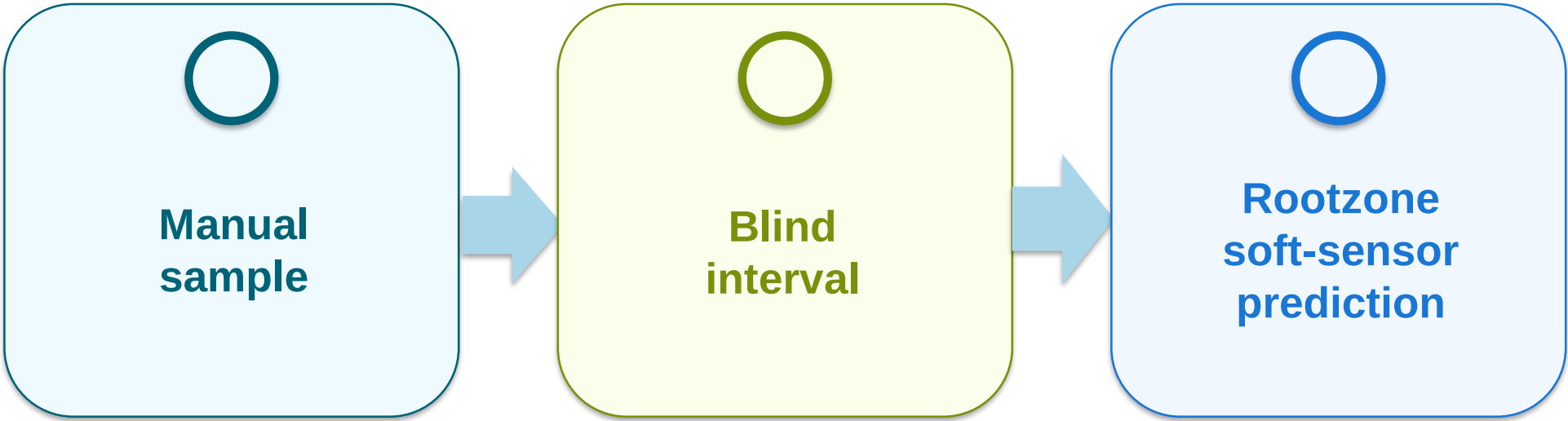
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1 Introduction & Goal

Why rootzone monitoring needs a soft sensor.

Rootzone pH and EC are key indicators of nutrient availability, salinity, and plant health in greenhouse cultivation. Manual samples are accurate, but they are sparse - between samples, acidity or salinity changes may go undetected.



Goal - Build a rootzone soft sensor that predicts future pH and EC between manual samples to optimize irrigation and fertigation decision-making.

2 Dataset & Processing

Five greenhouse domains synchronized into one 10-minute master dataset.

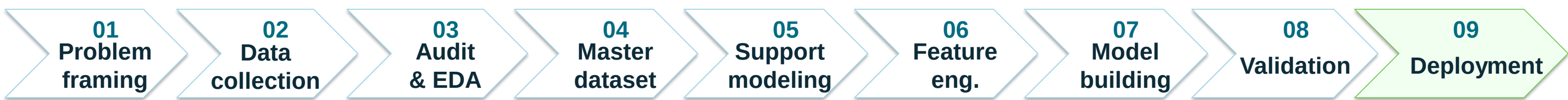
Climate conditions	temperature - humidity - radiation
Irrigation & fertigation	fertilizer - acid & salt inputs
Plant status	canopy cover - crop age - growth stage
Rootzone measurements	manual pH & EC samples
Time & history	timestamps - gaps - previous state

MASTER DATASET 10-minute greenhouse grid	
16,682 ROWS	10 min GRAIN
109 PH LABELS	109 EC LABELS
20 COLUMNS	May-Sep 2025

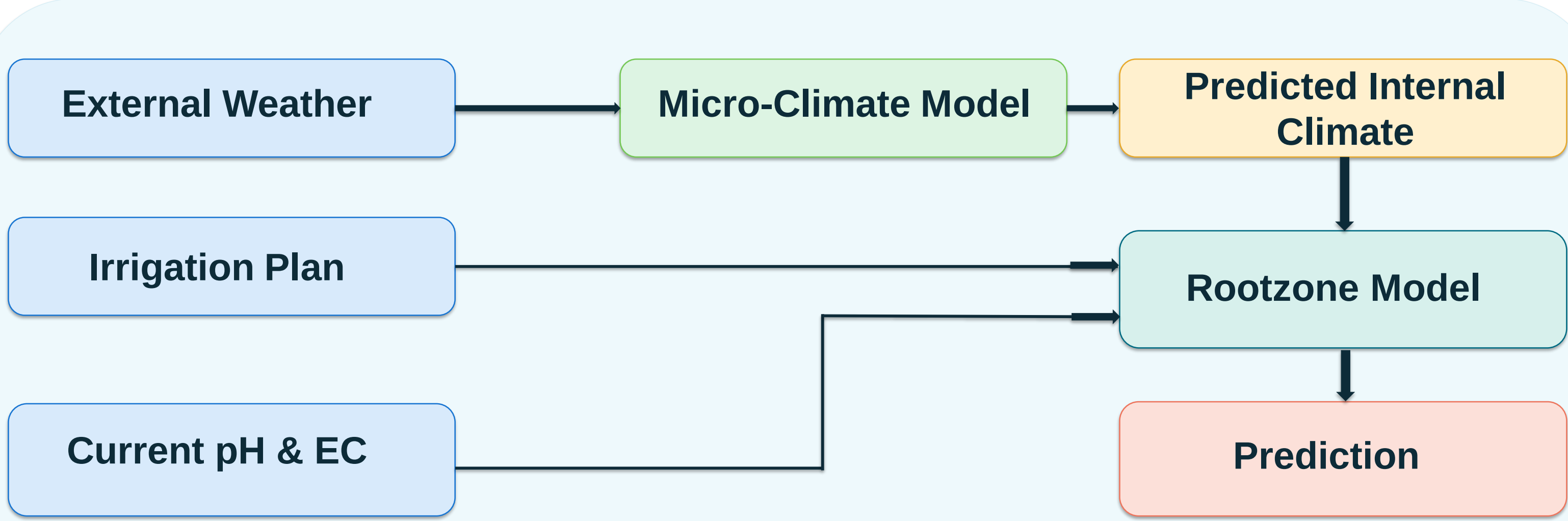
! Main data challenge: dense greenhouse telemetry, sparse rootzone labels.

3 System Architecture & Workflow

PROJECT WORKFLOW



SYSTEM ARCHITECTURE

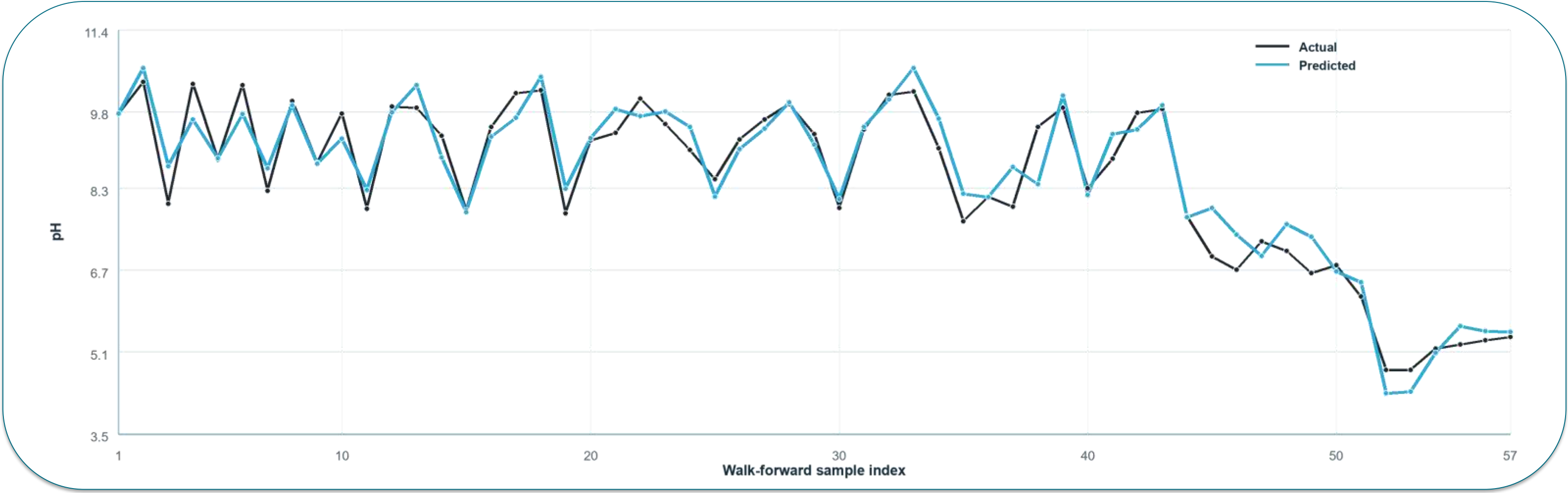


5 Results

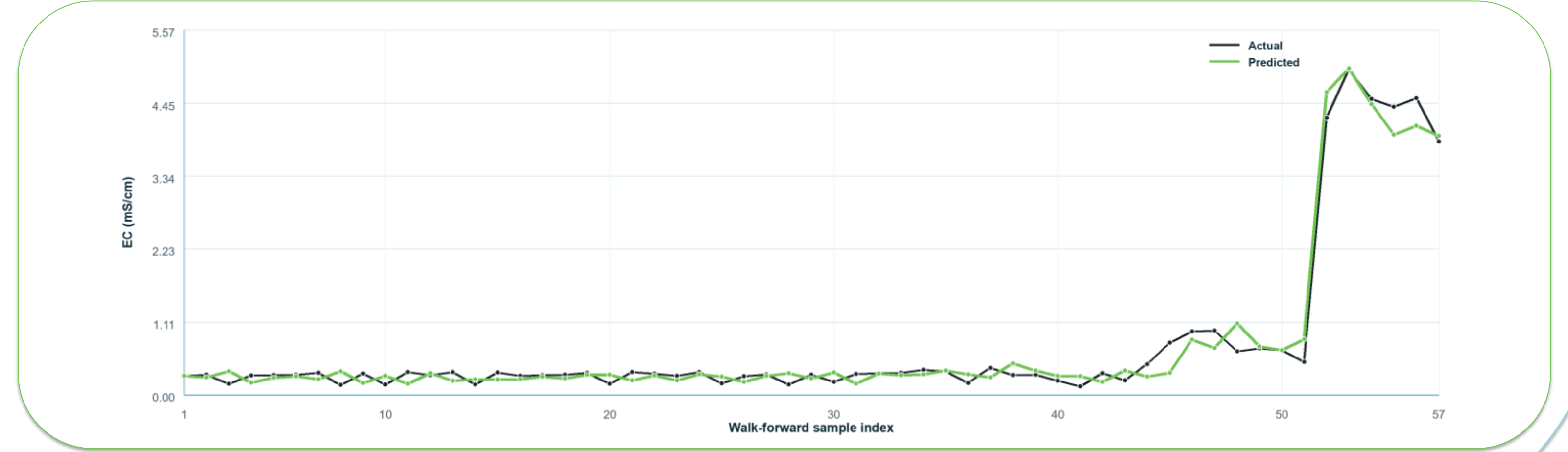
Wide walk-forward traces from export data: actual and predicted.

pH			EC		
0.330 MAE	0.932 R^2	62.4% GAIN	0.127 MAE	0.982 R^2	35.2% GAIN

pH ACTUAL VS PREDICTED - WALK-FORWARD



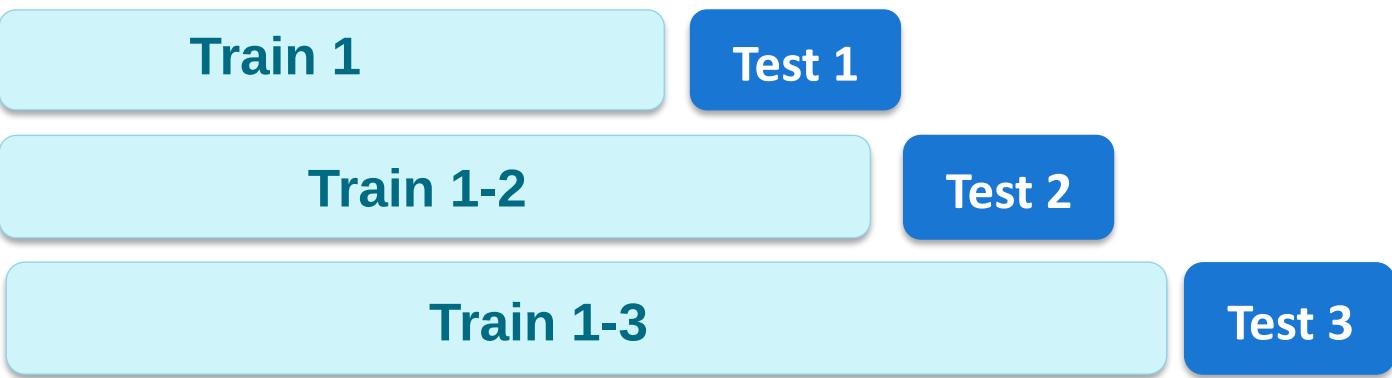
EC ACTUAL VS PREDICTED - WALK-FORWARD



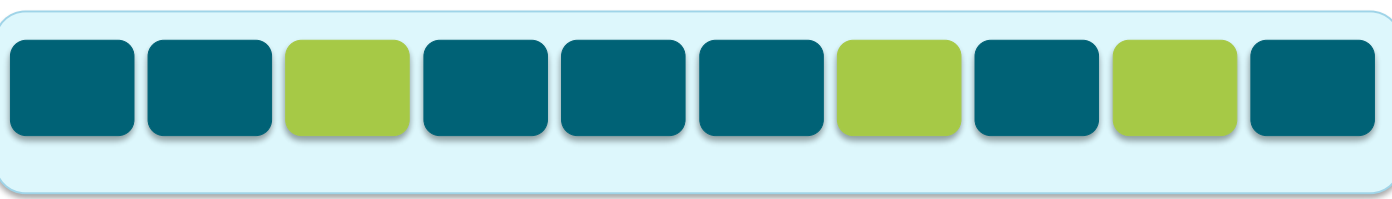
4 Model Development & Validation

Validation separates future testing, skipped intervals, and hybrid model behavior.

WALK-FORWARD STAIRCASE



SKIPPED-INTERVAL HOLDOUT



Hold out labeled intervals, then predict them later.

ROOTZONE HYBRID MODEL

Robust linear model Handles extrapolation beyond nearby manual samples.	Gradient boosted trees Interpolates local pH/EC movement between samples.
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Hybrid rootzone soft sensor
pH(t) + EC(t)

6 Conclusions & Impact

Sparse manual measurements become actionable guidance for irrigation and fertigation decisions.

- Earlier awareness**
Detect pH and EC movement between manual samples.
- Decision optimization**
Support irrigation and fertigation timing and dosing.
- Usable workflow**
Automated ETL, prediction, and dashboard packaging.